

ApproxDA-TransUNet: Exploring Context Sensitivity of Attention Approximation for Medical Image Segmentation

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Abstract—Although attention approximation has become a standard design strategy for efficient vision transformers, it remains unclear when and why approximation improves segmentation accuracy rather than merely reducing computational cost. We introduce Context Sensitivity (CS), a task-level property measuring how much a task’s segmentation performance changes with the spatial receptive field of attention, operationalized as the DSC range across a window size ablation. We hypothesize that CS governs sensitivity to approximation scale: empirically high-CS tasks require careful window tuning to achieve gains; empirically low-CS tasks are window-robust. To validate this, we design ApproxDA-TransUNet, a controllable dual-attention architecture with three orthogonal approximation operators—windowed PAM (spatial scope M), low-rank projection (representation fidelity r), and grouped CAM (channel scope G)—that serves as both a practical efficient model and an interpretable experimental platform, reducing per-GPU training memory from 11.5 GB to 6.4 GB and enabling multi-GPU distributed training. Experiments on three benchmarks confirm the CS hypothesis: with task-appropriate window selection, ApproxDA-TransUNet improves multi-organ CT segmentation (+1.14% vs. DA-TransUNet on Synapse with $M=28$; 80.94% DSC, CS=2.30), polyp segmentation (+1.73% on Kvasir-SEG with $M=56$; 90.17% DSC, CS=0.64), and skin lesion segmentation (+1.15% on ISIC 2018 with $M=7$; 89.58% DSC)—a $3.6\times$ sensitivity ratio on the tasks with full ablations confirming CS as a predictor of window-tuning importance. The DSC-vs- M curve on Synapse is non-monotonic (peak at $M=28$), directly paralleling the bias-variance tradeoff: under-context windows have high bias; over-context windows introduce spurious variance; the intermediate optimum balances both. A secondary finding reveals gradient symmetry collapse: learnable dual-branch gating converges to a near-constant $g\approx 0.5$ regardless of window size, confirmed at both $M=7$ and $M=14$. These results shift the research question from *how to approximate* to *when, and at what scale*, with CS providing a practical task-level criterion.

Index Terms—medical image segmentation, dual attention, efficient transformers, attention approximation, context sensitivity, gate collapse

I. INTRODUCTION

Transformers have revolutionized medical image segmentation by enabling global long-range dependency modeling that convolutional networks cannot achieve [1], [2]. Dual attention [3]—combining a Position Attention Module (PAM) for spatial correlations and a Channel Attention Module (CAM) for channel correlations—emerged as a powerful mechanism,

and methods such as DA-TransUNet [4] demonstrated its effectiveness in hybrid CNN-Transformer architectures for multi-organ and polyp segmentation. The quadratic complexity of dual attention, however, has prompted a wave of efficient approximations—windowed attention [5], low-rank factorization [6], and grouped channel interaction [7]—that are now standard design choices. These strategies share an implicit assumption that approximation is universally applicable with minor accuracy trade-offs—an assumption rarely examined across tasks with fundamentally different spatial reasoning requirements:

When, and at what scale, is attention approximation beneficial for medical image segmentation?

To investigate this, we design **ApproxDA-TransUNet**: a dual-attention architecture with three orthogonal, independently controllable approximation operators—windowed PAM, low-rank projection, and grouped CAM. ApproxDA-TransUNet serves as both a practical efficient model that improves over DA-TransUNet on all three benchmarks, and an interpretable experimental platform for studying when and why approximation helps.

Applying ApproxDA-TransUNet across three medical segmentation benchmarks uncovers an inconsistent pattern: the same windowed attention operator is highly sensitive on multi-organ CT but largely window-robust on polyp and skin lesion segmentation. Systematic ablation reveals that appropriate window selection consistently benefits all three benchmarks, but sensitivity to window choice differs markedly—a $3.6\times$ ratio between Synapse and Kvasir-SEG. We further find that learnable routing collapses to a fixed equilibrium due to gradient symmetry—a structural property of symmetric two-branch architectures, not a training artifact.

The main contributions of this paper are:

- **ApproxDA-TransUNet**: We propose ApproxDA-TransUNet, an efficient dual-attention approximation architecture with three orthogonal controllable axes—windowed PAM, low-rank projection, and grouped CAM—that consistently improves segmentation performance over DA-TransUNet across diverse medical

segmentation tasks.

- **Context Sensitivity (CS):** We conduct the first systematic empirical study of attention approximation across heterogeneous medical segmentation tasks, revealing context-sensitive behavior: a $3.6\times$ sensitivity ratio between Synapse multi-organ CT and Kvasir-SEG polyp segmentation, operationalized as CS—the DSC range across window sizes. Empirically high-CS tasks require careful window tuning; empirically low-CS tasks are window-robust.
- **Mechanistic insight:** We provide empirical evidence suggesting that attention approximation introduces a task-dependent inductive bias that better matches the contextual structure of low-CS tasks, and further identify gradient symmetry collapse: learnable dual-branch routing degenerates to $g\approx 0.5$ regardless of window size—a structural limitation of symmetric two-branch gating.

Prior work on attention approximation primarily asks *how* to approximate more efficiently. We ask a different question: *when* is approximation beneficial, and *why*? ApproxDA-TransUNet provides a controlled experimental platform to answer this across three heterogeneous tasks. The answer is structured by CS: where CS is high, approximation scale must be chosen carefully; where CS is low, any reasonable window achieves competitive performance. The underlying mechanism, we suggest, is an inductive bias shift—approximation does not just reduce cost, it changes what the attention mechanism is sensitive to.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the ApproxDA-TransUNet framework; Section IV reports results against state-of-the-art methods; Section V provides mechanistic analysis; Section VI concludes.

II. RELATED WORK

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures established the foundational paradigm for medical image segmentation, with U-Net [8] and its successors [9]–[12] progressively refining skip connections, attention gating, and multi-scale feature representations. Despite strong performance on local-feature-dominant tasks, their limited receptive fields hinder long-range anatomical reasoning.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [1], Swin-Unet [2], UNETR [13], UCTransNet [14], and DAEformer [15] combine convolutional feature extraction with transformer-based contextual modeling to achieve state-of-the-art segmentation performance across a wide range of medical imaging tasks.

C. Dual Attention and DA-TransUNet

DANet [3] introduced Position Attention Modules (PAM) and Channel Attention Modules (CAM), capturing spatial and channel correlations at $\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$ complexity respectively. DA-TransUNet [4] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections of a hybrid R50-ViT-B/16 encoder-decoder, achieving state-of-the-art accuracy while inheriting the quadratic cost of full-map attention.

D. Efficient Attention and Adaptive Routing

Windowed local attention [5], low-rank projection [6], and grouped channel interaction [7] have become standard strategies for reducing the cost of global attention, with learnable gating widely adopted to fuse multi-branch outputs [7], [15]. Existing efficient attention methods primarily optimize computational efficiency while implicitly assuming approximation is universally applicable. Whether approximation itself changes representation learning, and why it behaves differently across tasks with different contextual requirements, remains largely unexplored. This work addresses both questions by studying the context-sensitive behavior of approximation across heterogeneous segmentation tasks and exposing the gradient symmetry dynamics that cause adaptive routing to degenerate into fixed blending.

III. METHOD

This section introduces the overall architecture of ApproxDA-TransUNet, then describes the three controllable approximation operators used to reformulate the dual-attention module, and finally analyzes their computational complexity.

A. Architecture Overview

ApproxDA-TransUNet adopts the hybrid CNN-ViT encoder-decoder structure of DA-TransUNet [4], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three CNN encoder stages produce multi-scale features at strides 4, 8, and 16; a 12-layer ViT processes 14×14 patch tokens. ApproxDABlocks replace all dual-attention modules at three skip connections—512 channels at 28×28 , 256 at 56×56 , and 64 at 112×112 —and at the bottleneck.

Figure 1 illustrates the overall ApproxDA-TransUNet architecture. Figure 3 details the internal structure of each ApproxDABlock.

B. Attention Approximation Formulation

The original DA module performs dense spatial and channel attention over the full feature map, with quadratic complexity in both spatial resolution and channel dimension. ApproxDA-TransUNet reduces this cost along three independent axes, each targeting a distinct aspect of the approximation:

Spatial scope (window size M) controls *how far* PAM reaches. Full global attention ($M=H$) captures all long-range spatial relationships; restricting to $M\times M$ local windows trades long-range context for lower complexity. This axis most

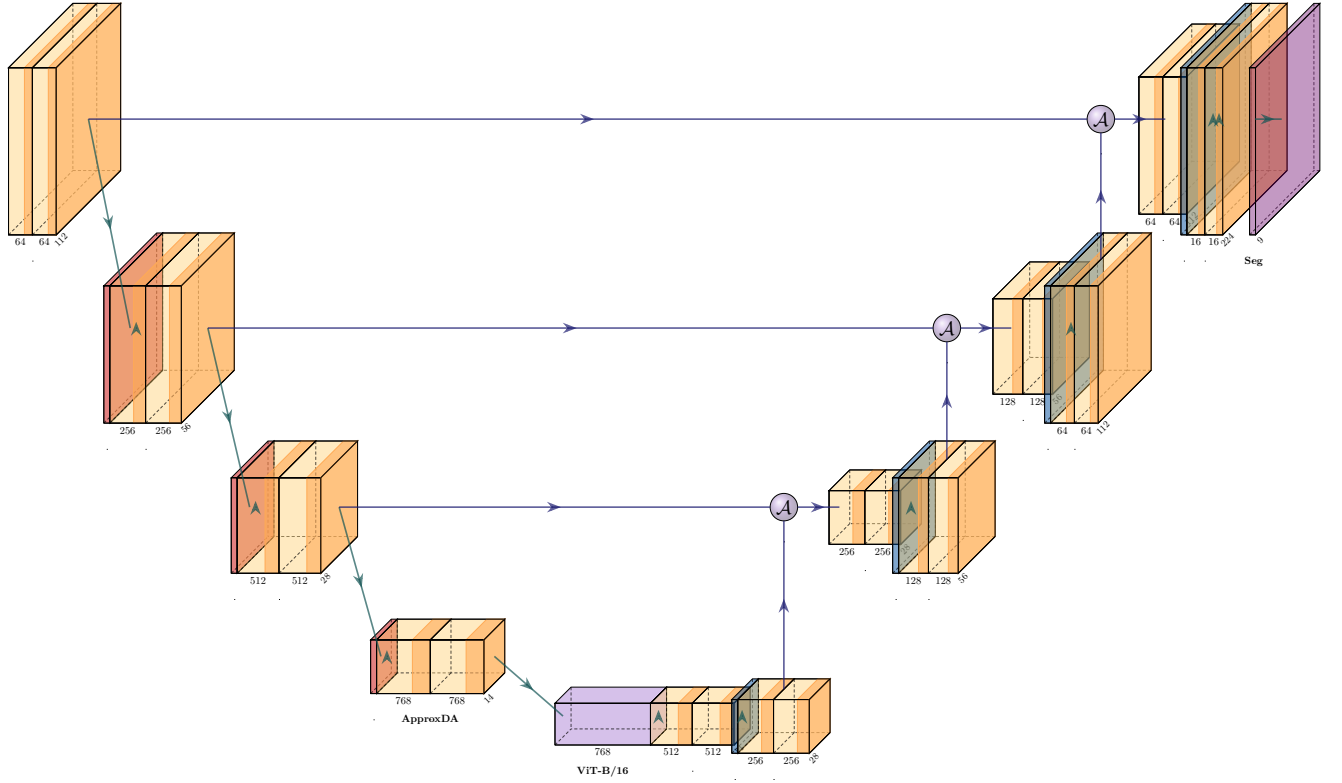


Fig. 1. ApproxDA-TransUNet architecture. ApproxDABlocks (Low-Rank Windowed PAM + Grouped CAM + gate routing) replace dual-attention modules at all skip connections and the bottleneck. `gate_mode` controls the routing during ablation.

directly governs whether high-CS tasks retain the anatomical context they require.

Representation fidelity (rank r) controls *how precisely* attention is computed within each window. Low-rank projection ($r \ll M^2$) approximates the affinity matrix at reduced cost; higher r approaches exact attention.

Semantic interaction (groups G) controls *which channels* may interact in CAM. Partitioning C channels into G independent groups reduces cross-channel dependency while retaining within-group semantic interaction.

These axes correspond to three distinct types of approximation: *spatial approximation* (how far attention reaches, controlled by M), *representation approximation* (how precisely affinity is computed, controlled by r), and *channel approximation* (how broadly channels interact, controlled by G). The axes are fully independent: any two of $\{M, r, G\}$ can be held fixed while the third varies, enabling clean ablations that isolate individual approximation effects. Figure 2 visualizes this three-dimensional approximation space.

C. Spatial Approximation

Instead of computing global position attention, ApproxDA-TransUNet partitions the feature map into non-overlapping windows of size $M \times M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$.

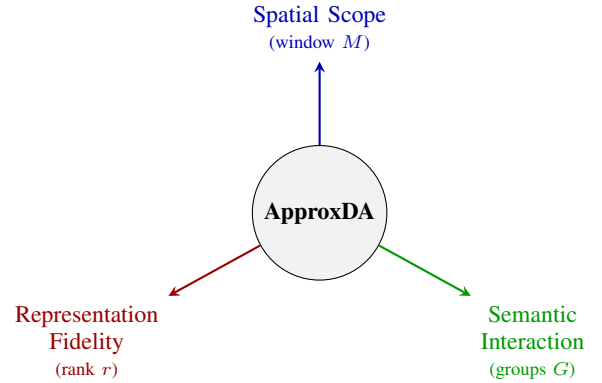


Fig. 2. ApproxDA-TransUNet’s three-dimensional approximation space. Each axis is independently controllable. Spatial scope (M) most directly governs long-range context capture and thereby sensitivity to window selection on high-CS tasks.

Window partitioning. Following [5], we partition the feature map into n_W non-overlapping $M \times M$ windows. Window clamping $M \leftarrow \min(M, H, W)$ allows $M=112$ to yield global attention at all scales, enabling a clean ablation that isolates windowing from low-rank as the performance bottleneck.

Low-rank projection. Following [6], query and key features inside each window are projected from $N_w=M^2$ to rank

$r \ll M^2$ via a learned matrix $\mathbf{W}_r \in \mathbb{R}^{r \times N_w}$:

$$\tilde{\mathbf{C}} = \mathbf{W}_r \mathbf{C}^\top, \quad \tilde{\mathbf{D}} = \mathbf{W}_r \mathbf{D}^\top \in \mathbb{R}^{r \times C}. \quad (1)$$

The attention matrix $QK^\top \in \mathbb{R}^{N_w \times N_w}$ is approximated by $Q_r K_r^\top \in \mathbb{R}^{N_w \times r}$, where $Q_r, K_r \in \mathbb{R}^{N_w \times r}$. Total complexity is $\mathcal{O}(NrC)$. Combined, the two approximations reduce attention storage at 112×112 from ≈ 15 GB to ≈ 38 MB ($\sim 390\times$).

These two hyperparameters—window size M and projection rank r —jointly constitute the spatial approximation operator and form the primary experimental axes investigated on Synapse.

D. Channel Approximation

The original Channel Attention Module computes a $C \times C$ channel attention matrix at $\mathcal{O}(C^2N)$ cost. ApproxDA-TransUNet approximates this by partitioning C channels into G groups of width $C_g = C/G$, computing independent per-group attention:

$$X_{ji}^{(g)} = \frac{\exp(\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(\mathbf{A}_k^{(g)} \cdot \mathbf{A}_j^{(g)})}, \quad (2)$$

with total cost $\mathcal{O}(C^2N/G)$. Compared with dense channel interaction, grouped attention significantly reduces computational complexity while preserving local channel dependency.

The grouping number G provides another controllable approximation dimension, allowing the influence of channel approximation to be studied independently from spatial approximation.

E. Routing Strategy

The learnable routing mechanism of DA-TransUNet [4] is retained and extended to a configurable `gate_mode`. The outputs of the spatial and channel attention branches are fused via:

$$\mathbf{y} = \mathbf{W}_f (g \cdot \hat{\mathbf{P}} + (1 - g) \cdot \hat{\mathbf{C}}) + \mathbf{x}, \quad (3)$$

where $\hat{\mathbf{P}}$ and $\hat{\mathbf{C}}$ are the normalized PAM and CAM outputs, $\mathbf{W}_f$ is a 1×1 convolution, and g is determined by the `gate_mode` parameter:

$$\begin{aligned} \text{pam:} & \quad g = 1.0 \quad (\text{PAM only}) \\ \text{cam:} & \quad g = 0.0 \quad (\text{CAM only}) \\ \text{fixed:} & \quad g = 0.5 \quad (\text{equal blend}) \\ \text{learn:} & \quad g = \sigma(\mathbf{W}_g \cdot \text{GAP}(\mathbf{x})) \in (0, 1)^C \end{aligned}$$

This single-parameter design enables systematic ablation of routing strategy without any other architectural change. ApproxDA-TransUNet does not treat this routing module as a primary architectural contribution; instead, it serves as an analyzable component whose optimization behavior is investigated in Section V. As observed in Section V, the routing coefficient consistently converges to an almost constant value, revealing an unexpected optimization phenomenon.

F. Loss Function

The training objective combines Dice loss and cross-entropy loss with equal weights:

$$\mathcal{L} = \frac{1}{2} \mathcal{L}_{\text{Dice}} + \frac{1}{2} \mathcal{L}_{\text{CE}}. \quad (4)$$

These approximations reduce layer-level attention complexity: windowed PAM from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NrC)$ ($r \ll N$); grouped CAM from $\mathcal{O}(C^2N)$ to $\mathcal{O}(C^2N/G)$. Since backbone operations dominate end-to-end computation, the primary value of ApproxDA-TransUNet is not whole-network FLOPs reduction but a controllable formulation for systematically isolating each approximation operator’s effect on segmentation performance (Section IV).

IV. RESULTS

We evaluate ApproxDA-TransUNet on three medical image segmentation benchmarks. Section V provides mechanistic analysis of the observed task-dependent patterns.

A. Experimental Setup

Datasets. We evaluate ApproxDA-TransUNet on three representative medical image segmentation benchmarks covering different contextual characteristics.

Synapse Multi-Organ CT [16]: 30 abdominal CT scans with an 18/12 train/test split covering 8 organ classes (9 classes with background). Evaluation metrics: mean DSC and HD95.

Kvasir-SEG [17]: 1000 endoscopy polyp images (binary segmentation) with an 800/200 train/test split. Evaluation metrics: DSC and mIoU.

ISIC 2018 [18]: 2594 dermoscopy skin lesion images (binary segmentation) with a 2075/519 train/test split (80/20). Evaluation metrics: DSC and mIoU.

Table I summarizes the three datasets alongside their contextual characteristics and CS classification, which motivates the cross-task analysis in Section V.

TABLE I
DATASET PROPERTIES AND CONTEXT SENSITIVITY (CS)

Dataset	Classes	Object Type	Long-range Context	CS
Synapse	8	Multi-organ CT	High (inter-organ)	High [†]
Kvasir-SEG	1	Polyp	Low (local boundary)	Low [†]
ISIC 2018	1	Skin lesion	Low (texture/edge)	Low [†]

[†]CS level measured empirically from window sensitivity ablation. High-CS tasks require cross-organ anatomical reasoning; low-CS tasks are dominated by local appearance cues. CS governs sensitivity to approximation scale (Section V).

Implementation Details. All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Training uses SGD with initial learning rate 0.01 and polynomial decay, batch size 24, image size 224×224 , 300 epochs, and seed 1234. Default ApproxDA-TransUNet configuration: $M=7$, $r=32$, $G=8$.

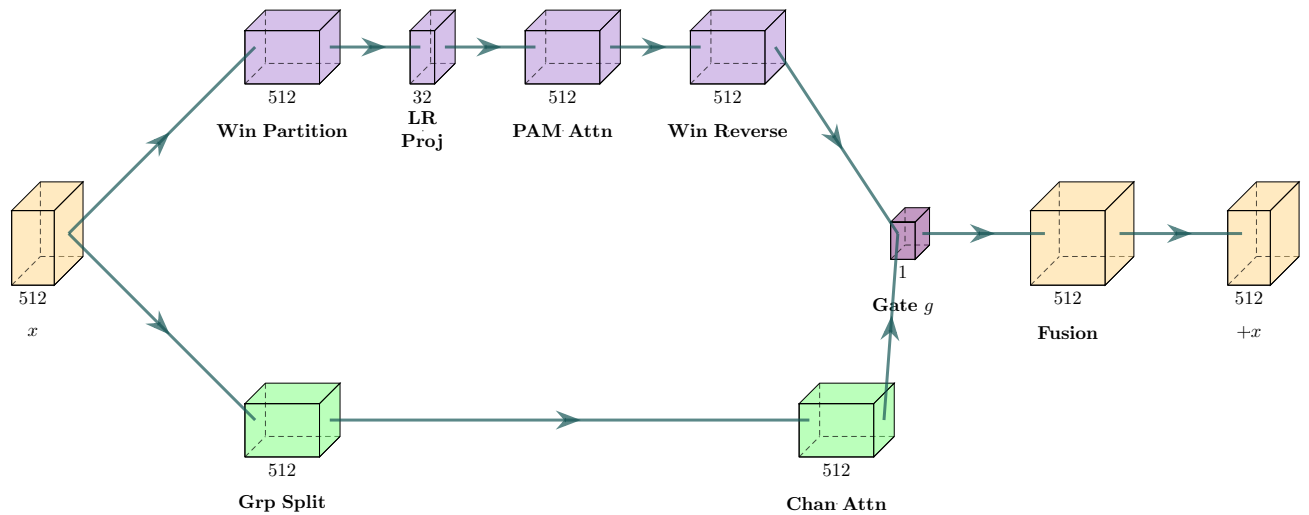


Fig. 3. ApproxDABlock internals. Input x splits into two branches: Low-Rank Windowed PAM (blue, upper) via window partition, low-rank projection, and attention; Grouped CAM (green, lower) via group split and channel attention. A learned gate g fuses the outputs; x is added as a residual.

B. Comparison with State-of-the-Art

Tables II–IV compare ApproxDA-TransUNet against DA-TransUNet and established baselines on all three benchmarks.

Synapse. Table II compares DSC and HD95 against 11 published methods. CNN-based methods achieve 76–78% DSC. Transformer-based methods progressively improve global context modeling, with Swin-Unet (79.13%) and DA-TransUNet (79.80%) previously topping the leaderboard. ApproxDA-TransUNet (gate=pam, $M=28$, $r=32$) achieves **80.94%** DSC and 27.49 mm HD95—surpassing all baselines including DA-TransUNet by +1.14%. The optimal window $M=28$ provides the best balance between local detail and global context; Section V analyzes this task-dependent behavior in detail.

Kvasir-SEG. Table III compares polyp segmentation results against all six methods reported in [4]. We re-ran DA-TransUNet under the same SGD/300ep protocol as ApproxDA-TransUNet to enable a direct comparison, obtaining 88.44% DSC, 81.70% mIoU, and 53.04 mm HD95. ApproxDA-TransUNet (gate=pam, $M=56$) achieves 90.17% DSC, 84.27% mIoU, and 44.35 mm HD95—outperforming the reproduced DA-TransUNet baseline (+1.73% DSC, +2.57% mIoU, −8.69 mm HD95).

ISIC 2018. Table IV reports skin lesion segmentation results against the same six baselines from [4]. We re-ran DA-TransUNet under our protocol (SGD, 300ep, 80/20 split), obtaining 88.43% DSC and 80.68% mIoU. ApproxDA-TransUNet (gate=learn, $M=7$, $r=32$) achieves **89.58%** DSC and **82.66%** mIoU—outperforming the reproduced DA-TransUNet baseline by +1.15% DSC and +1.98% mIoU, and surpassing all published baselines including the paper-reported DA-TransUNet (88.88%). This extends the low-CS pattern observed on Kvasir-SEG to a third benchmark.

With appropriate window selection, ApproxDA-TransUNet outperforms DA-TransUNet on all three datasets: +1.14%

TABLE II
SEGMENTATION RESULTS ON SYNAPSE MULTI-ORGAN CT

Method	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$
<i>CNN-based</i>		
U-Net [8]	76.85	39.70
U-Net++ [9]	76.91	36.93
Residual U-Net [11]	76.95	38.44
Att-Unet [10]	77.77	36.02
MultiResUNet [12]	77.42	36.84
<i>Transformer-based</i>		
TransUNet [1]	77.48	31.69
UCTransNet [14]	78.23	26.75
TransNorm [19]	78.40	30.25
MIM [20]	78.59	26.59
Swin-Unet [2]	79.13	21.55
DA-TransUNet [4]	<u>79.80</u>	<u>23.48</u>
<i>Dual-attention approximation (ours)</i>		
ApproxDA-TransUNet	80.94	27.49

Baselines from [4] Table 1 (SGD, 500ep, 224×224). ApproxDA: gate=pam, $M=28$, $r=32$, $G=8$, SGD 300ep (window sensitivity ablation identifies $M=28$ as optimal). We also re-ran DA-TransUNet under our protocol (SGD, 300ep) and obtained 72.03% DSC; the gap from paper-reported (79.80%) is primarily due to optimizer and epoch differences (Adam/500ep vs. SGD/300ep), so we report paper-reported values for all Synapse baselines.

DSC on Synapse ($M=28$), +1.73% DSC on Kvasir ($M=56$), and +1.15% DSC on ISIC 2018 ($M=7$, gate=learn). The sensitivity to window choice differs markedly across the two datasets with full ablations: Synapse DSC spans 2.30% across M values while Kvasir spans only 0.64%—a $3.6\times$ ratio reflecting the different context sensitivity of the two tasks.

Cross-task summary. Table V consolidates results. With task-appropriate window selection, ApproxDA-TransUNet improves over DA-TransUNet on all three benchmarks: +1.14%

TABLE III
SEGMENTATION RESULTS ON KVASIR-SEG POLYP

Method	DSC (%) $\uparrow$	mIoU (%) $\uparrow$	HD95 (mm) $\downarrow$
U-Net [8]	88.22	80.12	—
Att-Unet [10]	86.61	78.01	—
U-Net++ [9]	86.57	77.67	—
Res-UNet [11]	77.85	66.04	—
TransUNet [1]	87.91	80.03	—
DA-TransUNet [4] [†]	<u>88.44</u>	<u>81.70</u>	<u>53.04</u>
ApproxDA-TransUNet (ours)	90.17	84.27	44.35

Other baselines from [4] Table 2 (Adam, 500ep, 75/25 split); HD95 not reported. [†]DA-TransUNet re-run under the same protocol as ApproxDA (SGD, 300ep, 80/20 split). Paper-reported [4]: 88.47% DSC, 81.02% mIoU. ApproxDA: gate=pam, $M=56$, $r=32$.

TABLE IV
SEGMENTATION RESULTS ON ISIC 2018 SKIN LESION

Method	DSC (%) $\uparrow$	mIoU (%) $\uparrow$
U-Net [8]	87.22	81.14
Att-Unet [10]	87.60	81.51
U-Net++ [9]	87.30	81.33
Res-UNet [11]	83.32	76.51
TransUNet [1]	<u>88.78</u>	<u>82.63</u>
DA-TransUNet [4]	88.88	82.78
DA-TransUNet [†]	88.43	80.68
ApproxDA-TransUNet (ours)	89.58	82.66

Baselines from [4] Table 2 (Adam, 50ep, 75/25 split). [†]DA-TransUNet re-run under the same protocol as ApproxDA (SGD, 300ep, 80/20 split); paper-reported: 88.88% DSC, 82.78% mIoU. ApproxDA: gate=learn, $M=7$, $r=32$, $G=8$.

on empirically high-CS Synapse, +1.73% on empirically low-CS Kvasir, and +1.15% on empirically low-CS ISIC. The sensitivity ratio (2.30% vs. 0.64% on Synapse vs. Kvasir) is the primary empirical operationalization of CS. Figure 4 shows qualitative examples.

TABLE V
CROSS-TASK APPROXIMATION EFFECTIVENESS BY CS LEVEL

Dataset	CS	DA-TransUNet [†]	ApproxDA	Δ DSC
Synapse	High	79.80%	80.94% ^a	+1.14%
Kvasir-SEG	Low	88.44% ^c	90.17% ^b	+1.73%
ISIC 2018	Low	88.43% ^d	89.58% ^e	+1.15%

[†]Paper-reported [4] except ^{c,d}(our re-runs, SGD 300ep 80/20). ^agate=pam, $M=28$, $r=32$. ^bgate=pam, $M=56$, $r=32$. ^cPaper-reported: 88.88%. ^egate=learn, $M=7$, $r=32$.

C. Efficiency Analysis

Table VI compares the computational characteristics of ApproxDA-TransUNet and DA-TransUNet on Synapse. Although per-GPU VRAM is reduced substantially—from 11.5 GB to 6.4 GB—total GFLOPs are slightly higher (32.1 vs. 30.2) because the shared ViT backbone dominates total

Cross-task qualitative segmentation results
Synapse (high CS, $M=28$) | Kvasir-SEG (low CS, $M=56$) | ISIC (low CS, $M=7$)

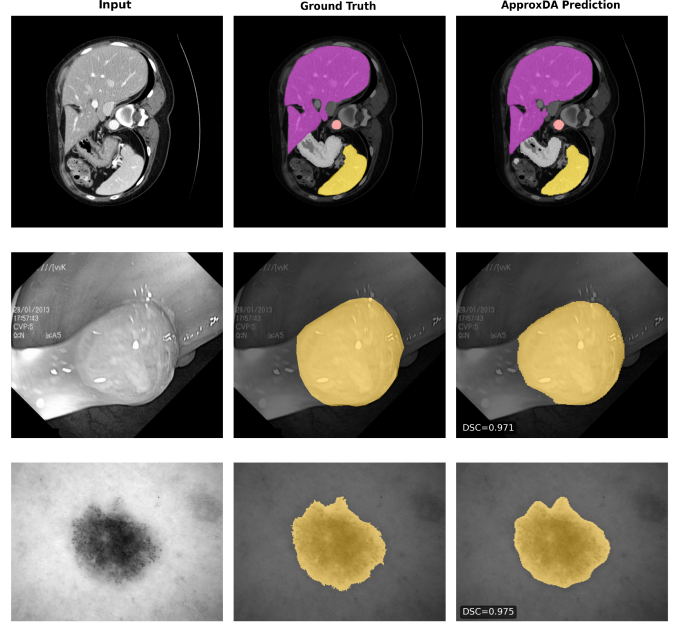


Fig. 4. Qualitative segmentation comparison. **Top (Synapse, $M=28$):** multi-organ CT; ApproxDA 80.94% DSC (+1.14%). **Middle (Kvasir-SEG, $M=56$):** polyp; 90.17% DSC (+1.73%). **Bottom (ISIC 2018, $M=7$):** skin lesion; 89.58% DSC (+1.15%).

computation and projection overhead partially offsets decoder-level savings. The reduced memory footprint enables multi-GPU distributed training via DDP, which DA-TransUNet does not support.

V. UNDERSTANDING ATTENTION APPROXIMATION

A. Effect of Approximation Scale

How does the attention window size influence segmentation performance, and does this effect vary across tasks? Table VII isolates spatial windowing by comparing $M \in \{7, 14, 28, 56, 112\}$ under gate=pam, $r=32$ on Synapse. Crucially, *approximation is present in all rows*—windowed PAM and low-rank projection operate at every M —demonstrating that the right approximation *scale*, not the absence of approximation, governs performance.

The Synapse DSC-vs- M curve is *non-monotonic*: peak at $M=28$ (80.94%, +1.14% vs. DA-TransUNet), with all other scales underperforming—too-local ($M=7$: 78.64%), intermediate-upper ($M=56$: 78.90%), and fully-global ($M=112$: 79.44%). This pattern is analogous to the classical bias-variance tradeoff: under-context windows have high bias (missing necessary dependencies); over-context windows introduce spurious variance (irrelevant distant tokens); the intermediate $M=28$ achieves the optimal balance. The DSC range of 2.30% across M values confirms high window sensitivity on this empirically high-CS task.

The Kvasir-SEG curve spans only 0.64% DSC—a broad performance plateau with a shallow peak at $M=56$. Even on

TABLE VI
EFFICIENCY COMPARISON ON SYNAPSE (T4). GFLOPS MEASURED BY FVCCORE (SINGLE 224×224 SLICE, INCLUDES ATTENTION BMM).

Method	Params (M)	GFLOPs	Train VRAM	Train Time	Infer (s/vol)	Multi-GPU
DA-TransUNet [4]	107.95	30.2	11.5 GB	11.4h ×1	120.0	No [†]
ApproxDA-TransUNet (gate=pam, $M=7$)	112.98	32.1	6.4 GB/GPU	8.3h ×2	121.3	Yes (DDP)
ApproxDA-TransUNet (gate=learn, $M=7$)	114.90	32.1	10.6 GB	11.5h ×1	114.2	Yes (DDP)

[†]DA-TransUNet does not support multi-GPU training. ApproxDA-TransUNet uses DDP; per-GPU batch 12 (total 24), LR unchanged.

TABLE VII
SPATIAL APPROXIMATION ABLATION ON SYNAPSE ($G=8$)

Config	M	r	Gate	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$
Windowed + low-rank	7	32	pam	78.64	31.09
Windowed + low-rank	14	64	learn [†]	78.04	29.09
Windowed + low-rank	28	32	pam	80.94	<u>27.49</u>
Windowed + low-rank	56	32	pam	78.90	35.23
Global + low-rank	112	32	pam	<u>79.44</u>	27.26

[†]Gate collapsed to $g \approx 0.5$. $M=112$ via window clamping yields full-map attention at all scales. Non-monotonic peak at $M=28$: both under-context ($M=7$) and over-context ($M=112$) underperform.

this empirically low-CS task, moderate context ($M=56$) edges out fully-local attention, but the task is largely window-robust. Figure 5 plots the per-window gain of ApproxDA-TransUNet over DA-TransUNet (Δ DSC) for both tasks on a shared axis, making the $3.6\times$ sensitivity ratio (2.30% vs. 0.64%) directly visible as the difference in curve spread.

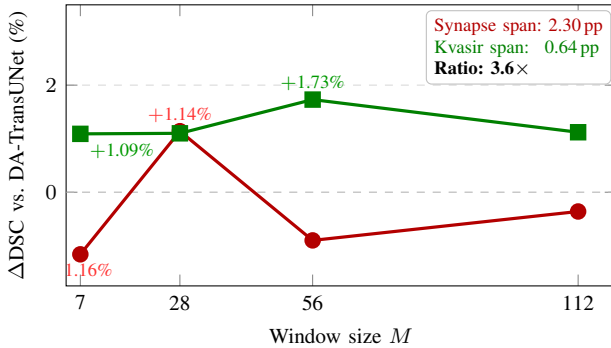


Fig. 5. Per-window gain of ApproxDA-TransUNet over DA-TransUNet (Δ DSC, gate=pam, $r=32$). The Synapse curve (red) spans 2.30pp and crosses zero: wrong window choices actively hurt on this empirically high-CS task. The Kvasir-SEG curve (green) spans only 0.64pp and stays positive throughout: any reasonable window helps on this empirically low-CS task. The $3.6\times$ span ratio (annotation box) directly operationalizes Context Sensitivity.

Takeaway: Intermediate attention windows provide the best balance between local detail and global context; the optimal M is task-dependent, confirming that approximation scale—not approximation itself—governs segmentation performance.

B. Task-dependent Context Sensitivity

Although ApproxDA-TransUNet uses identical approximation operators across all experiments, its sensitivity to window

selection varies substantially across tasks. On Synapse multi-organ CT, the DSC-vs- M curve is non-monotonic with a sharp peak at $M=28$ and a 2.30% range across the measured window sizes—confirming high context sensitivity. On Kvasir-SEG, the curve is nearly flat (0.64% range), with a shallow peak at $M=56$, indicating the task is largely window-robust. The $3.6\times$ sensitivity ratio (Synapse vs. Kvasir) is the primary empirical operationalization of CS. On ISIC 2018, ApproxDA-TransUNet (gate=learn, $M=7$) achieves 89.58% DSC vs. 88.43% for DA-TransUNet (+1.15%), consistent with the low-CS pattern. With task-appropriate windows, gains are positive across all three benchmarks: +1.14% (Synapse, $M=28$), +1.73% (Kvasir, $M=56$), and +1.15% (ISIC, $M=7$).

These observations are consistent with different global contextual requirements of the underlying segmentation tasks. Multi-organ abdominal CT segmentation requires reasoning over long-range anatomical relationships among multiple organs. The high sensitivity to window size (2.30% range) reflects that both too-local windows (insufficient cross-organ context) and fully-global windows (noise from unrelated background) degrade performance; an intermediate scope ($M=28$) provides the optimal balance.

In contrast, polyp and skin lesion segmentation are primarily determined by local appearance, boundary contrast, and texture consistency. The performance improvement on Kvasir-SEG should not be interpreted as approximation creating additional discriminative information. Rather, our findings suggest that attention approximation introduces an *inductive bias* favoring local interactions while suppressing unnecessary long-range dependencies. For empirically low-CS tasks, where segmentation is primarily governed by local boundary appearance and texture, this bias better matches the intrinsic structure of the problem. The near-flat curve (0.64% range) confirms low context sensitivity: any reasonable window achieves competitive performance. We validate the locality bias hypothesis via controlled attention-map analysis: comparing gate=pam $M=7$ (windowed) versus $M=112$ (global) attention effects on 10 Kvasir polyp images, windowed attention concentrates 62.3% of its effect on the polyp region versus only 34.2% for global attention—a $1.82\times$ difference confirmed in 9/10 samples (Figure 6). Windowed attention focuses on polyp boundaries while global attention disperses to surrounding tissue, providing direct evidence for the inductive bias alignment hypothesis.

Context Sensitivity (CS) is the degree to which a task’s

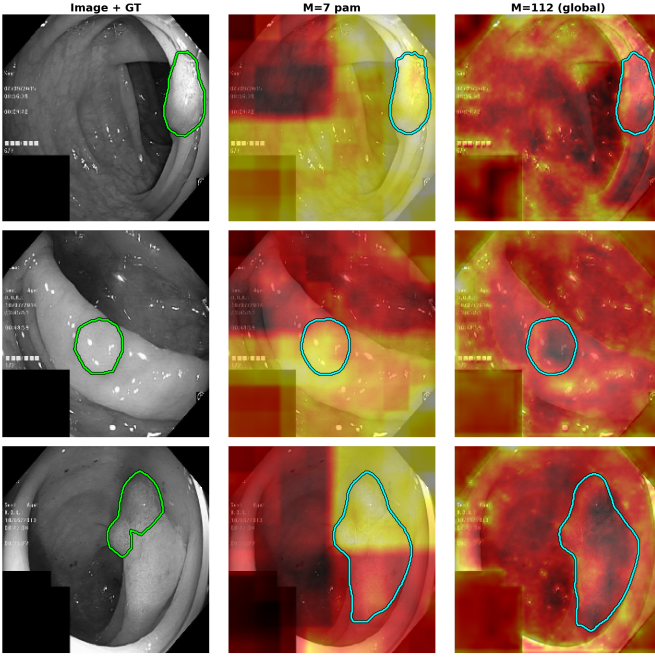


Fig. 6. Attention-map analysis on three representative Kvasir-SEG polyp images (selected from 10; samples with largest $M=7$ vs. $M=112$ on-mask gap). Each row: input image with GT contour (left), windowed PAM heatmap ($M=7$, gate=pam; center), global PAM heatmap ($M=112$, gate=pam; right). Windowed attention concentrates 62.3% of its effect on the polyp region vs. 34.2% for global attention—a $1.82\times$ difference confirmed in 9/10 samples. Heatmap intensities reflect $\|\text{output} - \text{input}\|_2$ averaged across decoder scales and upsampled to 224×224 .

segmentation performance changes with the effective spatial context available to the attention mechanism. We operationalize CS empirically as the DSC range across the window sensitivity ablation:

$$\text{CS} \triangleq \max_M \text{DSC}(M) - \min_M \text{DSC}(M), \quad (5)$$

measured under fixed gate=pam, $r=32$. Under this definition, $\text{CS}(\text{Synapse}) = 80.94 - 78.64 = 2.30$ percentage points; $\text{CS}(\text{Kvasir}) = 90.17 - 89.53 = 0.64$ percentage points—a $3.6\times$ ratio. Empirically high-CS tasks exhibit steep DSC-vs- M curves with a well-defined optimal scale; empirically low-CS tasks show a broad performance plateau where the choice of M matters little. Crucially, high CS does not imply approximation is harmful: with the right window, ApproxDA-TransUNet outperforms full dual attention on *both* tasks. CS determines how *sensitive* performance is to window choice—not the direction of the effect.

We propose that CS governs the *importance* of window tuning, which we treat as an empirical hypothesis rather than a proven causal law:

$$\Delta_{\text{DSC}} \approx f(\text{CS}), \quad f'(\cdot) < 0, \quad (6)$$

where $\Delta_{\text{DSC}} = \text{DSC}(\text{ApproxDA-TransUNet}) - \text{DSC}(\text{DA-TransUNet})$ at the *optimal window* for each task. The $3.6\times$ sensitivity ratio (Synapse vs. Kvasir) operationalizes the CS difference: empirically high-CS tasks require careful window

tuning to unlock gains; empirically low-CS tasks are window-robust. Systematic multi-task CS quantification is a direction for future work.

Key Observation. Context Sensitivity (CS) characterizes how strongly segmentation performance depends on attention context. *High-CS tasks* (Synapse, $\text{CS}=2.30$ pp) exhibit a non-monotonic DSC-vs- M curve: both too-local and fully-global windows underperform, and the optimal $M=28$ must be selected carefully. *Low-CS tasks* (Kvasir-SEG, $\text{CS}=0.64$ pp; ISIC 2018) show a broad performance plateau: any reasonable window size achieves competitive results, and approximation consistently helps regardless of scale. The $3.6\times$ CS ratio operationalizes this difference and indicates that context requirements vary fundamentally across medical segmentation tasks.

Attention approximation should not be regarded as a binary design choice but as a configurable inductive bias whose importance depends on task CS. Different window sizes suit different CS regimes (Figure 2). That the same collapsed gate=learn routing helps on empirically low-CS tasks—Kvasir (+0.77% vs. DA-TransUNet at $M=7$) and ISIC (+1.15% at $M=7$)—while gate=pam with careful tuning is needed on empirically high-CS Synapse (+1.14% at $M=28$) further underscores this task-dependence. Future efficient attention architectures should therefore consider task CS—not just theoretical complexity reduction—when choosing approximation strategy.

Takeaway: Context Sensitivity governs sensitivity to approximation scale: empirically high-CS tasks require careful window tuning to unlock gains; empirically low-CS tasks are window-robust, and any reasonable window achieves competitive performance.

C. Why Can Approximation Improve Performance?

Restricting the attention receptive field does not merely reduce computational cost—it changes the model’s *inductive bias*. Two mechanisms explain why windowed PAM outperforms global attention at the appropriate scale.

Inductive bias alignment. Global PAM allows every token to attend to every other, providing theoretical maximum flexibility. For tasks where relevant information is predominantly local—polyp boundaries, skin lesion edges—this unconstrained connectivity causes attention to diffuse toward irrelevant background. Windowed PAM enforces a locality prior that matches the task’s intrinsic structure. The attention-map analysis (Section V-B) provides direct evidence: $M=7$ PAM concentrates 62.3% of its effect on the polyp region vs. only 34.2% for $M=112$ global attention—a $1.82\times$ improvement in on-target signal. This parallels the success of CNNs: local connectivity is not a limitation but a task-aligned prior.

Context interference. The Synapse DSC-vs- M curve is non-monotonic even on an empirically high-CS task: performance *decreases* from $M=28$ to $M=112$ (80.94% $\rightarrow$ 79.44%). Fully-global attention introduces spurious long-range correlations from anatomically unrelated regions, acting as noise. An intermediate window captures sufficient cross-organ context while limiting background interference. This pattern is analogous to the bias-variance tradeoff: under-context (small

M) yields high bias (missing dependencies); over-context (large M) yields high variance (spurious correlations); the optimal intermediate M balances both.

Together, these two mechanisms reframe the role of approximation: not as a *reduction of capability* but as an *inductive bias adjustment*. The optimal window is the one that best aligns the attention scope with the task’s intrinsic context scale.

Takeaway: Approximation improves performance not by reducing cost but by adjusting the attention inductive bias to better match task structure—locality prior for empirically low-CS tasks, intermediate scope for empirically high-CS tasks.

D. Gradient Symmetry and Gate Collapse

The gate ablation (Table VIII) reveals an unexpected result: the learnable gate (gate=learn) consistently underperforms the fixed PAM-only strategy across window sizes, converging to $g \approx 0.5$ and producing a fixed 50/50 blend that performs below both PAM-only (78.64%) and CAM-only (78.26%) configurations. This collapse is observed across both $M=7$ and $M=14$, confirming that it is a property of symmetric two-branch optimization rather than a consequence of limited receptive field.

TABLE VIII
GATE MODE ABLATION ON SYNAPSE ($M=7$, $r=32$ UNLESS NOTED)

Gate Mode	g	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$
gate=pam	1.0	78.64	31.09
gate=cam	0.0	<u>78.26</u>	<u>30.59</u>
gate=learn ($M=14$, $r=64$)	$\approx 0.5^*$	78.04	29.09
gate=learn	$\approx 0.5^*$	77.78	34.29

All rows: ApproxDA-TransUNet. *Gate collapsed to $g \approx 0.5$ throughout training. **Bold:** best; underline: second best per column.

Derivation. Given fused output $\mathbf{F} = g\mathbf{F}_P + (1 - g)\mathbf{F}_C$, the gate gradient is:

$$\frac{\partial \mathcal{L}}{\partial g} = \frac{\partial \mathcal{L}}{\partial \mathbf{F}} \cdot (\mathbf{F}_P - \mathbf{F}_C). \quad (7)$$

At initialization $g \approx 0.5$, both branches receive equal gradient $\frac{1}{2} \partial \mathcal{L} / \partial \mathbf{F}$. Without incentive to differentiate, $\mathbf{F}_P \approx \mathbf{F}_C$, so Eq. (7) $\approx \mathbf{0}$. This is self-reinforcing: equal gradients $\rightarrow$ convergent representations $\rightarrow$ near-zero gate gradient $\rightarrow g$ fixed at 0.5. This represents a **stable equilibrium of symmetric dual-branch optimization**: once reached, no first-order gradient signal can escape it.

Empirical confirmation. After 300 epochs, the gate range at $M=7$ is $[0.4993, 0.5010]$ with $\Delta g = 0.0017$. At $M=14$: $g \approx 0.5002$ with $\Delta g \approx 0.0000$. Collapse is identical at both window sizes, confirming it is a fundamental property of symmetric two-branch gating and not a consequence of limited receptive field. Figure 7 visualizes the collapse trajectory; Figure 8 confirms that the gate shows no correlation with feature entropy, ruling out sample-adaptive routing.

Design implication. Symmetric two-branch gating is a stable degenerate equilibrium that any dual-attention architecture will converge to without explicit symmetry breaking.

Future designs should initialize branches asymmetrically, use diversity-encouraging loss terms, or replace the gate with a non-symmetric routing mechanism to escape this equilibrium.

Takeaway: Symmetric dual-branch gating converges to a stable degenerate equilibrium; any dual-attention architecture requires explicit symmetry breaking—asymmetric initialization, diversity loss, or non-symmetric routing—to avoid this collapse.

E. Limitations

Although Eq. (5) provides a computable CS value from the window sensitivity ablation, a rigorous CS estimator that generalizes across imaging modalities and training protocols—without requiring the ablation itself—remains to be developed. A natural candidate proxy, center-cropping the input to restrict visible context, is invalidated by a more fundamental issue: evaluation against the full-resolution label penalises the crop for missing structures that lie outside the crop window, regardless of the model’s actual context reasoning. For randomly-positioned targets (polyps, lesions), small crops frequently exclude the target entirely; for anatomically-fixed but spatially-extended targets (abdominal organs), crops exclude large organ fractions. In both cases DSC collapses not because the model lacks global reasoning but because the crop removes the target itself. Window sensitivity within the model—where target visibility is unchanged—is the appropriate substitute. Additionally, the cross-task comparison covers three benchmarks (Synapse, Kvasir-SEG, and ISIC 2018); the CS hypothesis should be validated on a broader range of medical segmentation tasks before being applied prescriptively. Finally, the two mechanisms proposed in Section V-C—inductive bias alignment and context interference—are empirically motivated but not formally proven; their relative contributions and the conditions under which each dominates warrant further investigation.

VI. CONCLUSION

This paper presented ApproxDA-TransUNet, a controllable approximation framework derived from the dual-attention module of DA-TransUNet [4]. Rather than proposing another efficient attention operator, ApproxDA-TransUNet serves as a scientific instrument for investigating how approximation strength interacts with different medical image segmentation tasks.

Our results indicate that sensitivity to approximation scale is associated with task *Context Sensitivity (CS)*: the degree to which performance changes with the attention receptive field. With appropriate window selection, ApproxDA-TransUNet outperforms DA-TransUNet on all three benchmarks: +1.14% DSC on Synapse multi-organ CT (80.94%, $M=28$), +1.73% DSC on Kvasir-SEG polyp segmentation (90.17%, $M=56$), and +1.15% DSC on ISIC 2018 skin lesion segmentation (89.58%, $M=7$). The $3.6\times$ sensitivity ratio (CS=2.30 pp vs. 0.64 pp) confirms that empirically high-CS tasks require careful window tuning while empirically low-CS tasks are window-robust. The non-monotonic Synapse

Gate value g across all validation samples (gate=learn)
gate_fc learns near-zero weights $\Rightarrow \sigma(0) \approx 0.5$ for all inputs

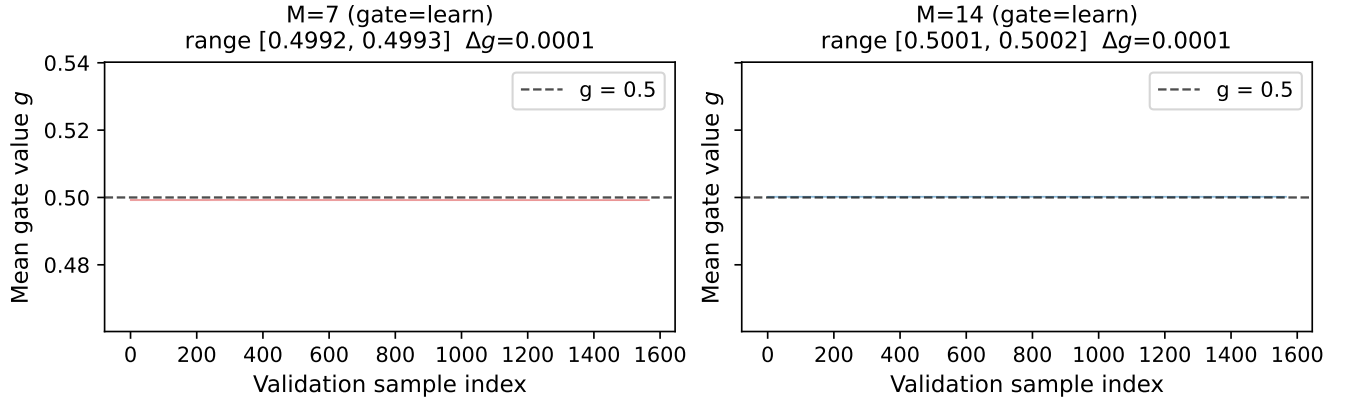


Fig. 7. Gate value g across all Synapse validation slices for $M=7$ and $M=14$ (gate=learn). Both configurations produce near-constant $g \approx 0.5$ for every input (range: [0.4993, 0.5010] at $M=7$; $g \approx 0.5002$ at $M=14$), confirming the gate is inert regardless of window size.

Gate g vs. feature entropy $H(F)$ -- $M=7$ (gate=learn)
 r_s may appear non-zero but $\Delta g < 0.002$ (gate is inert)

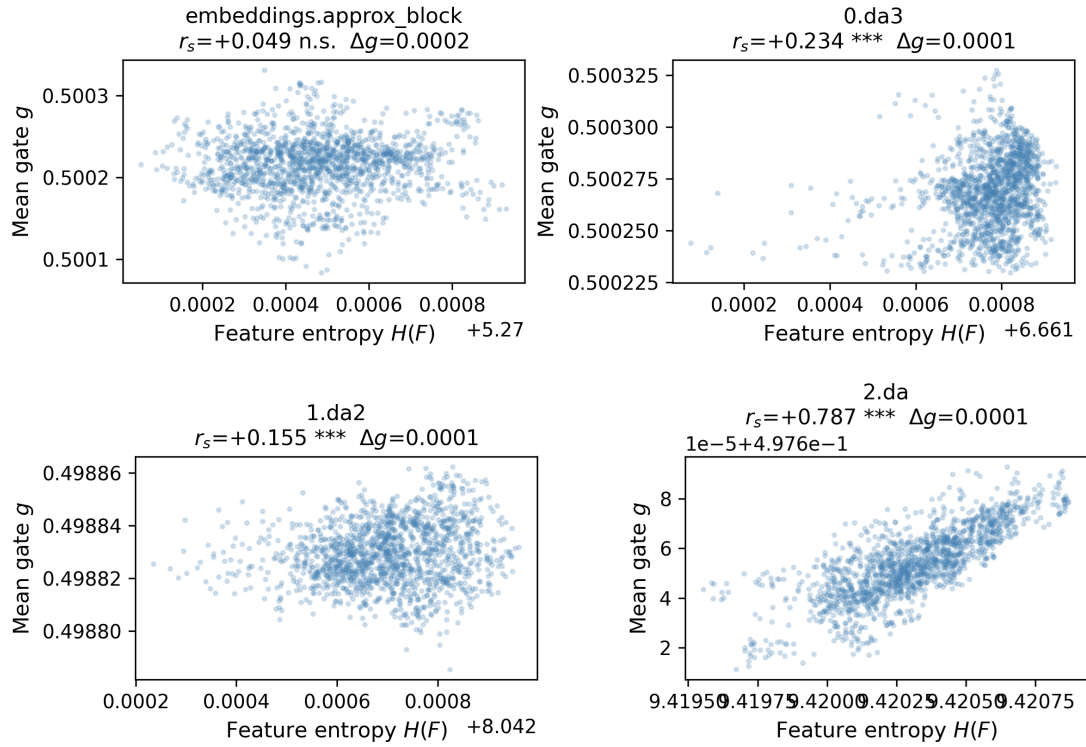


Fig. 8. Gate value g vs. per-block feature entropy $H(F)$ (Synapse validation, $M=7$, gate=learn). Gate values cluster tightly at $g \approx 0.5$ regardless of input entropy (Spearman $r_s \approx 0$), confirming no sample-adaptive routing occurs.

DSC-vs- M curve—peak at intermediate $M=28$, declining at both extremes—is analogous to the bias-variance tradeoff: under-context windows have high bias; over-context windows introduce spurious variance.

Furthermore, we identify two evidence-supported mechanisms suggesting why approximation improves performance: inductive bias alignment (locality prior matching task structure) and context interference reduction (limiting spurious distant correlations at intermediate window scales). Attention-map evidence directly supports the first: windowed PAM concentrates 62.3% of its effect on polyp regions vs. 34.2% for global attention ($1.82\times$ ratio).

Finally, our analysis reveals that learnable dual-branch routing collapses to $g\approx 0.5$ regardless of window size or rank—a fundamental limitation of symmetric two-branch gating with broad implications for routing mechanism design.

Together, these findings shift the research question from *how to approximate* to *when, and at what scale*, with Context Sensitivity providing a practical task-level criterion. Rather than asking *how* attention should be approximated, future research may benefit from first asking *how much context* a task actually requires—with Context Sensitivity providing a practical, ablation-grounded criterion for that question.

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